

FRDM-IMXRT1186

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1. Unless Otherwise Specified:
- All resistors are in ohms, 1/16 Watt,0402
 - All capacitors are in uF,0402
 - All voltages are DC
 - All polarized capacitors are aluminum electrolytic
2. Interrupted lines coded with the same letter or letter combinations are electrically connected.

Revision History

Rev. Code	Date	By	Description
A	2025-03-14	Shawn Shi	Initial Release
A1	2025-05-26	Shawn Shi	1.Update DC_5V_IN and Ethernet RJ45 VCC pin. 2.Replace HyperFlash with S26HL5121FPBHM000.
A2	2025-09-05	Shawn Shi	1.Replace RGMII PHY (U21, U29) with YT8531SH-CA. 2.Populate R243 and R448. 3.Add test point for POR_GTL, CCM.CLKO1 and CCM.CLKO2. 4.Replace CAN with TJA1057BT. 5.Replace R119 and R502 with 10ohm, replace R487 with 33ohm based on prescan test. 6.Add an inverter and pull up resistor for ECAT_RESET_OUT 7.Add test point for automated test platform.
A3	2025-10-21	Shawn Shi	DNP R30 and R31, added Arduino interface I2C note on page18. No layout change.
A4	2025-12-02	Shawn Shi	No layout change 1. Correct typo in the block diagram. 2. Change J4 from 1-2 to 5-6. 3. Improve statement boot switch default set up.

Jumper Setting

REF DES	JUMPER(DEFAULT)	PAGE NAME
J5	1-2	03 Main Power
J4	5-6	03 Main Power
J10	1-2	04 Power Domain
J15,J16,J19,J20,J21,J38	1-2	05 MIMXRT1180 Part1
J11,J37	Open	05 MIMXRT1180 Part1
J27,J28,J29,J30,J31,J32,J33, J35,J41,J43,J44,J45,J46,J47, J48	1-2	06 MIMXRT1180 Part2
J42,J49	2-3	06 MIMXRT1180 Part2
J12,J13,J17,J18	2-3	07 MIMXRT1180 Part3
J8,J24,J26	1-2	16 Boot & Reset
J7,J9,J14,J25	Open	16 Boot & Reset
J34,J59	1-2	17 MCU-LINK
J36,J40,J50,J58	Open	17 MCU-LINK
J55	Open	18 Arduino/JTAG/SRAMC

Switch Setting

REF DES	SWITCH(DEFAULT)	PAGE NAME
J60	pff.off, on	16 Boot & Reset

3. Device type number is for reference only. The number varies with the manufacturer.
4. Special signal usage:
- _B Denotes - Active-Low Signal
 - <> or [] Denotes - Vectored Signals
5. Interpret diagram in accordance with American National Standards Institute specifications, current revision, with the exception of logic block symbology.



Classification: <FCP> <PUB>

Drawing Title: FRDM-IMXRT1186

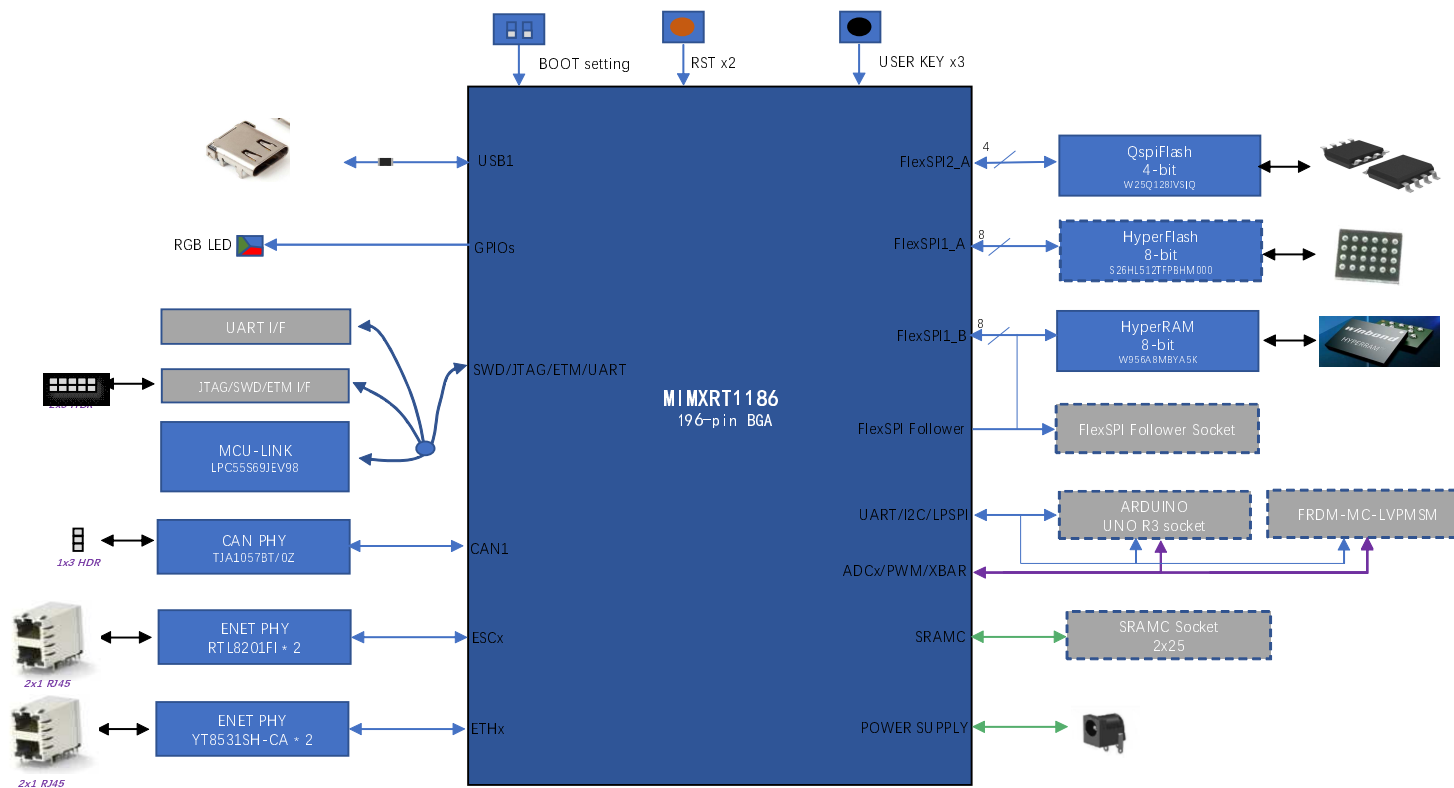
Page Title: Cover

Size C Document Number SCH-95302, PDF-SPF-95302

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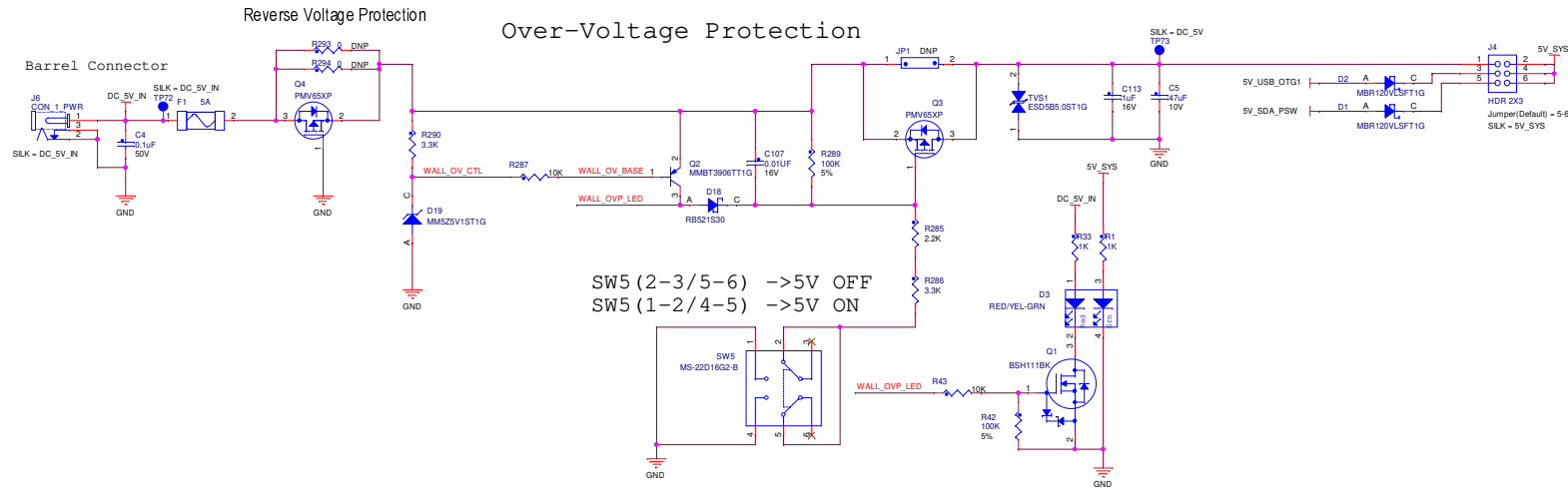
Rev A4

FRDM-IMXRT1186

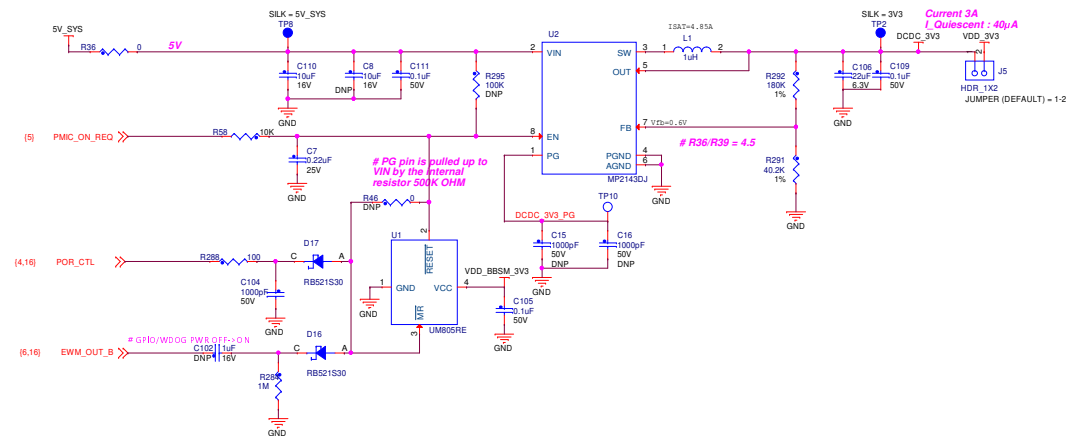


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Page Title:	Block Diagram	
Size	Document Number	Rev
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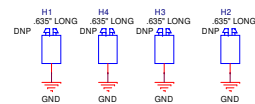
Main Power



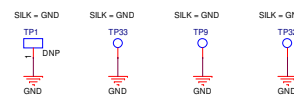
5V To 3.3V



Board Mounting Holes



Ground TPs

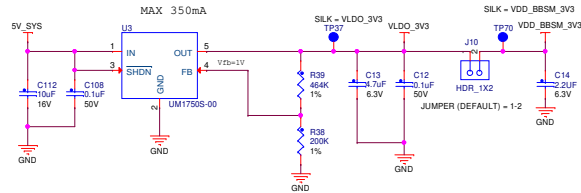


Layout Note: Place Ground TPs to assist signal measurement.

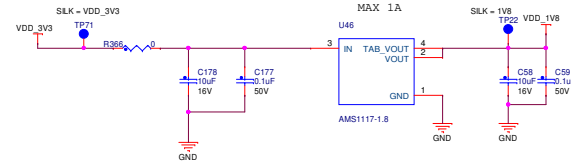


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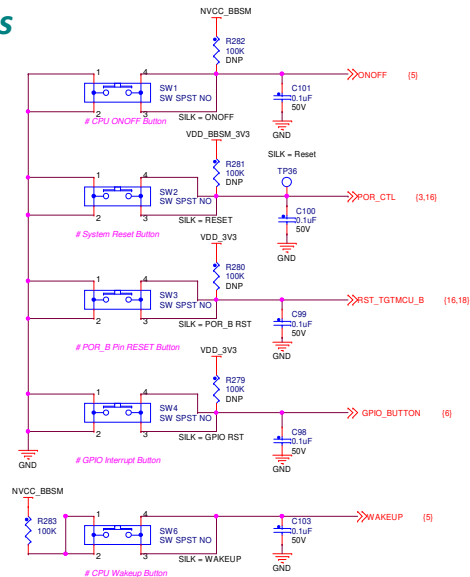
3V3 LDO for BBSM



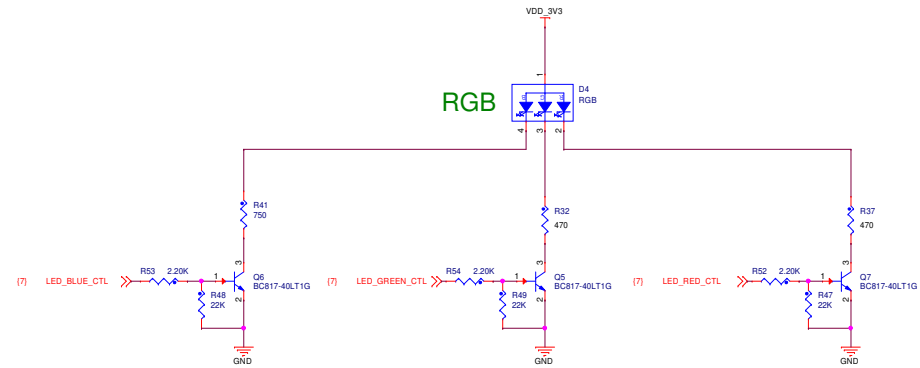
1V8 LDO



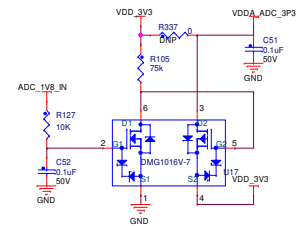
Buttons



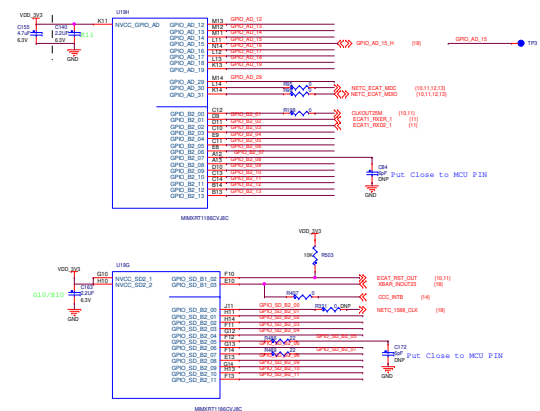
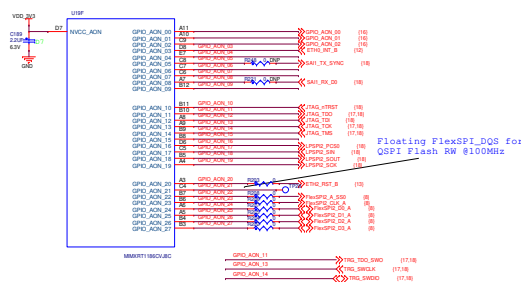
User LED



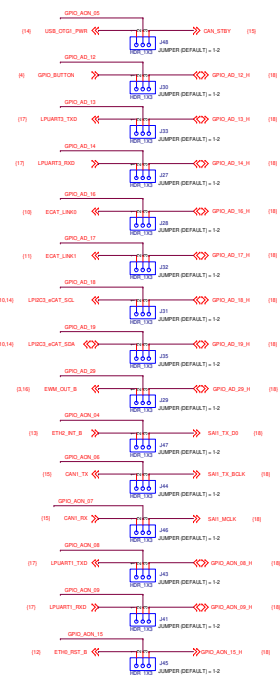
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Size	Document Number	Rev
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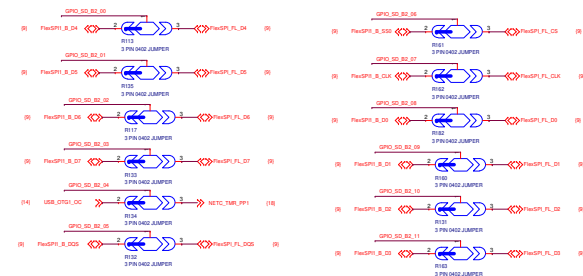
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Page Title: IMMXRT1180 Part1			
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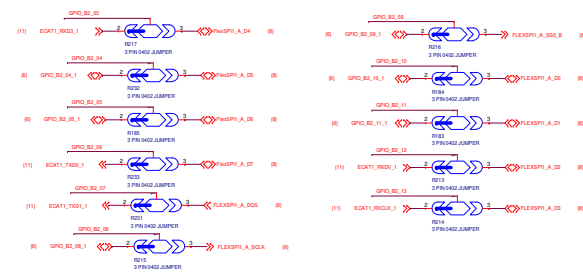
Function PinMux



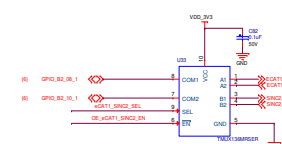
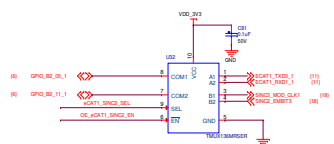
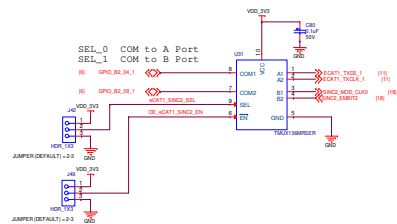
FlexSPI Follower Option(Reserved)

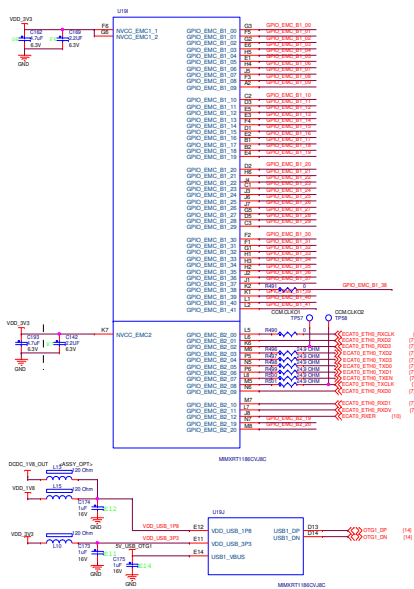


FlexSPI1 HyperFlash Option(Reserved)

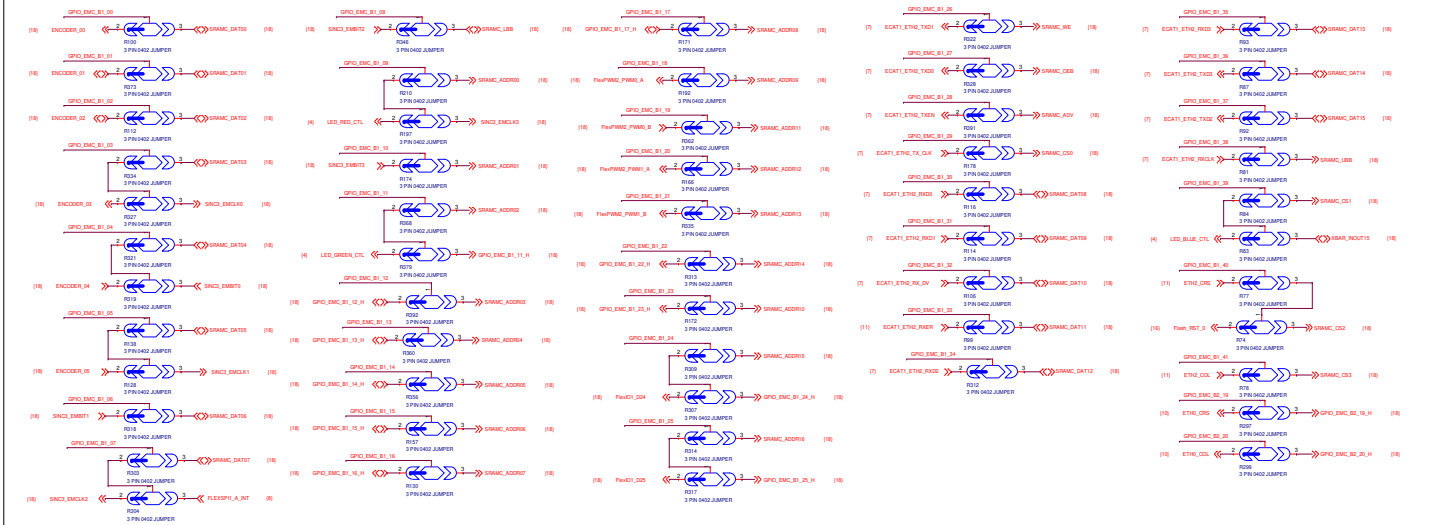


J42(2-3)/J49(2-3): eCAT Function Pin (Default)
J42(1-2)/J49(2-3): SINC Function Pin

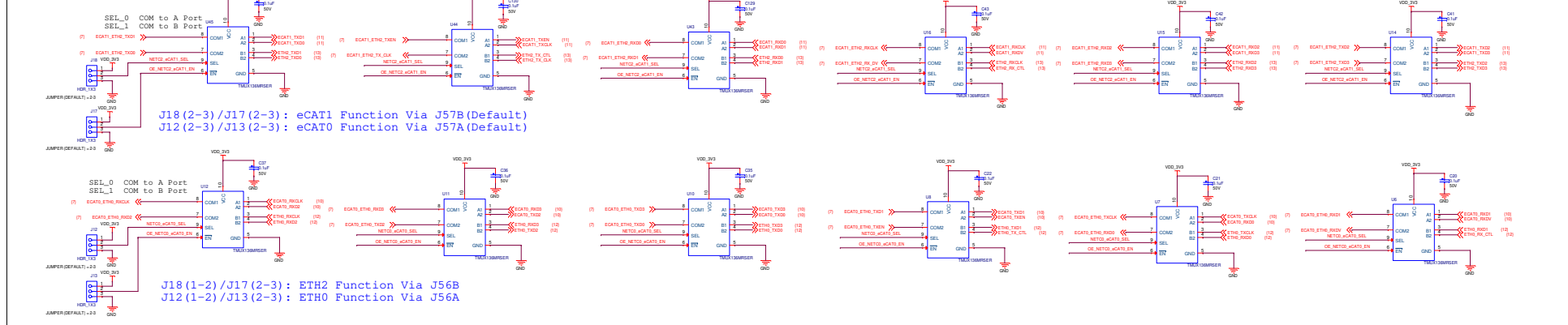




SRAMC Option(Reserved)

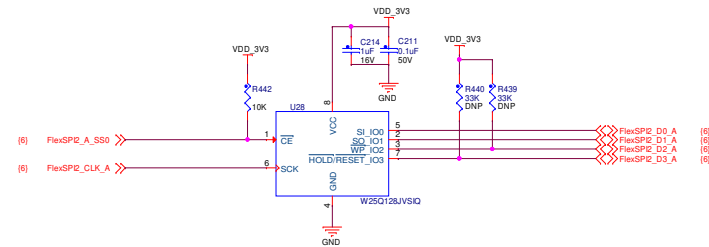


ECAT/ETH Option



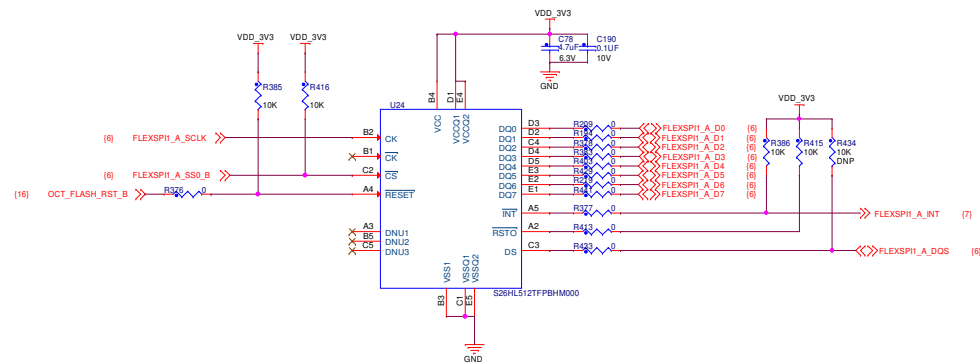
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Part Name:	FRDM-MXRT1180		
Page Title:	MXRT1180 Part3		
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QSPI Flash



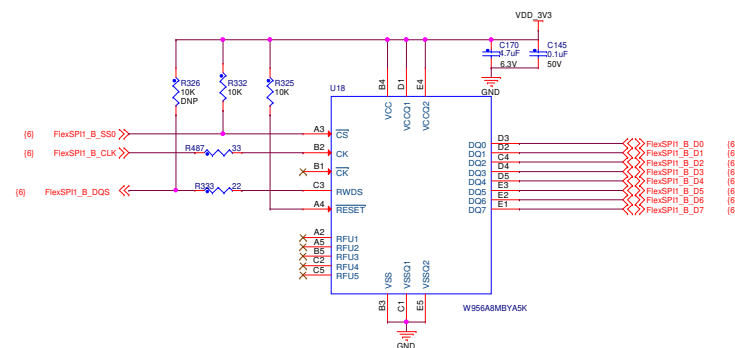
HyperFlash(Reserved)

To Enable HyperFlash:
R183~R185, R213~R217, R231~R233 Change to 1-3



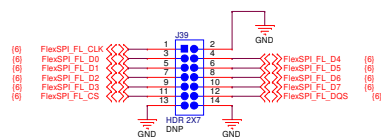
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HyperRAM



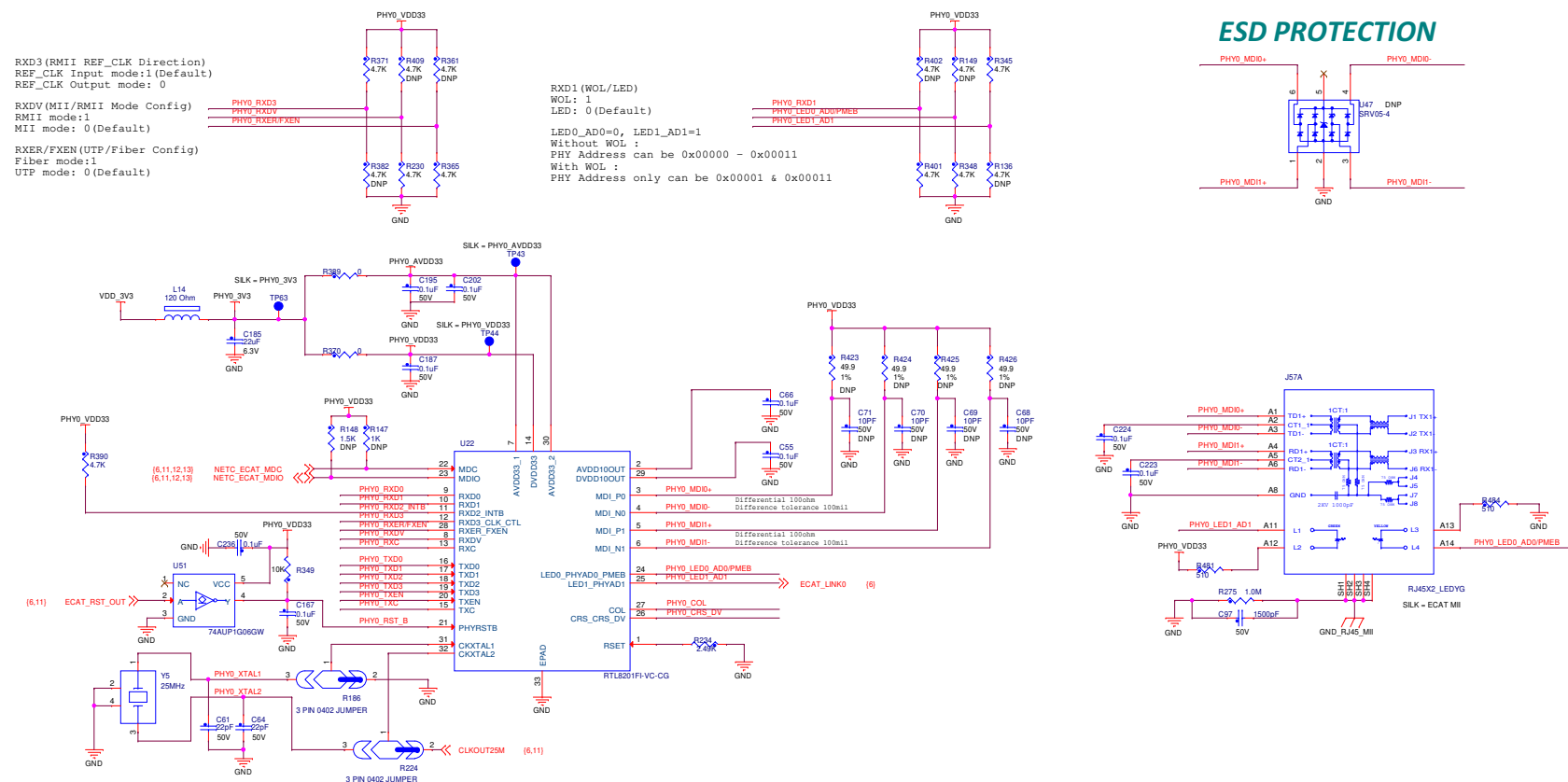
FlexSPI Follower(Reserved)

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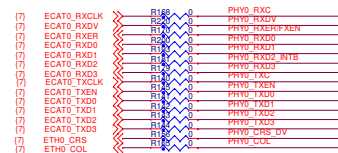


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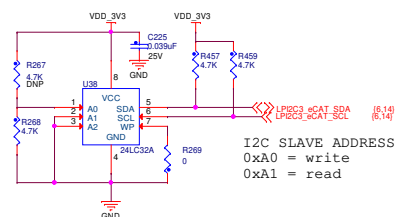
MII ECAT0 {J12(2-3)/J13(2-3): eCAT0 Function Via J57A(Default)}



MII Interface



EEPROM



				
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FRDM-IMXRT1186				
Part Title:				
MIL_ECATO				
Size C	Document Number:	SCH-95302, PDF-SPP-95302		Rev A4
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MII ECAT1 {J18(2-3)/J17(2-3): eCAT1 Function Via J28B(Default)}

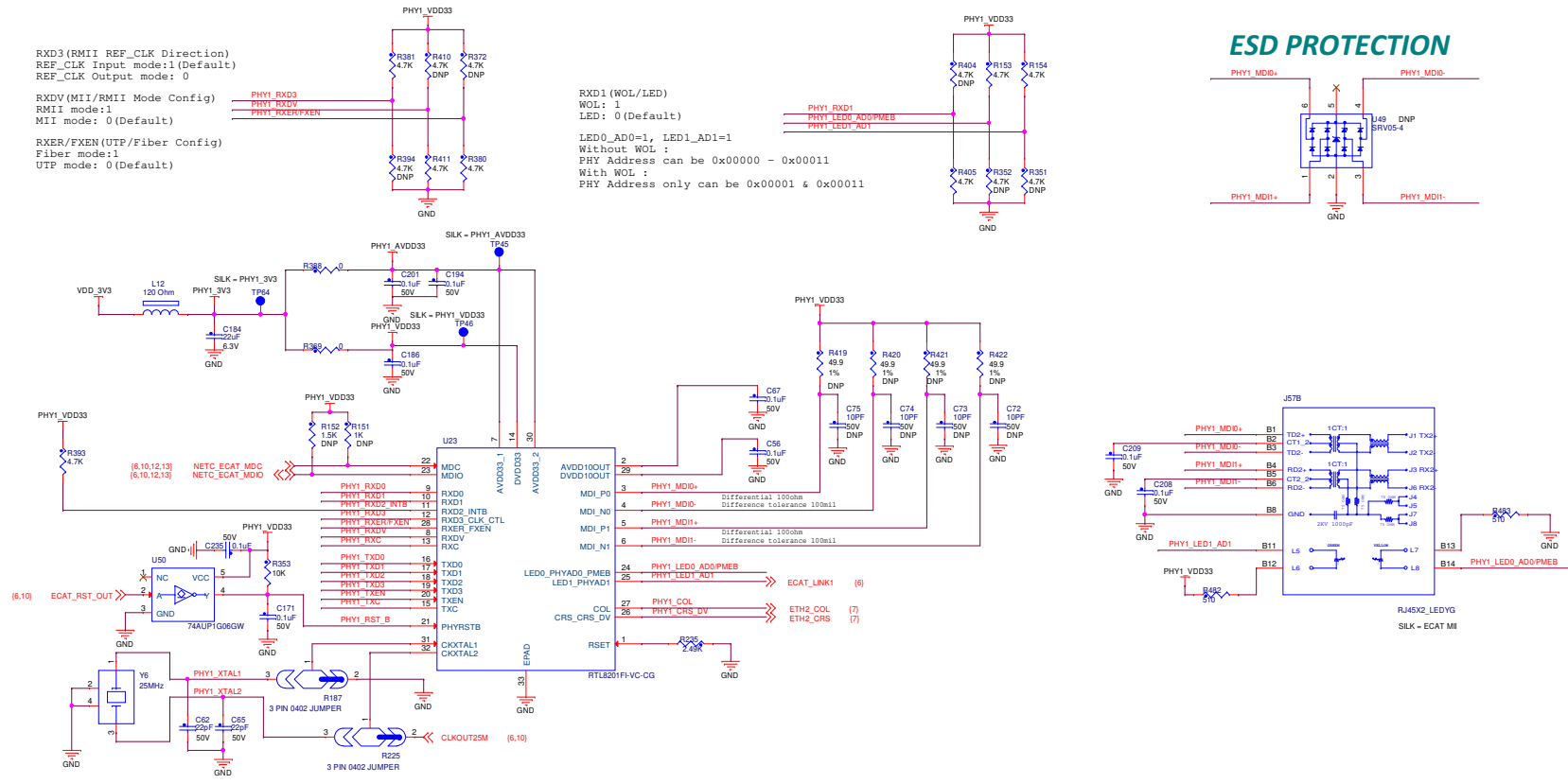
```
RXD3(RMII REF_CLK Direction)
REF_CLK Input mode:1(Default)
REF_CLK Output mode: 0
```

```
RXDV(MII/RMII Mode Config)
RMII mode:1
MII mode: 0(Default)
```

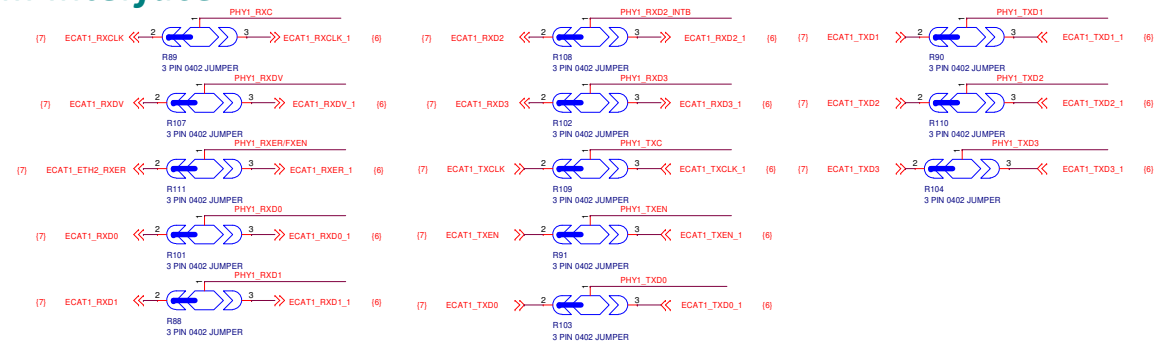
```
RXER/FXEN(UTP/Fiber Config)
Fiber mode:1
UTP mode: 0(Default)
```

```
RXD1 (WOL/LED)
WOL: 1
LED: 0 (Default)
```

```
LED0_AD0=1, LED1_AD1=1
Without WOL :
PHY Address can be 0x00000 - 0x00011
With WOL :
PHY Address only can be 0x00001 & 0x00011
```

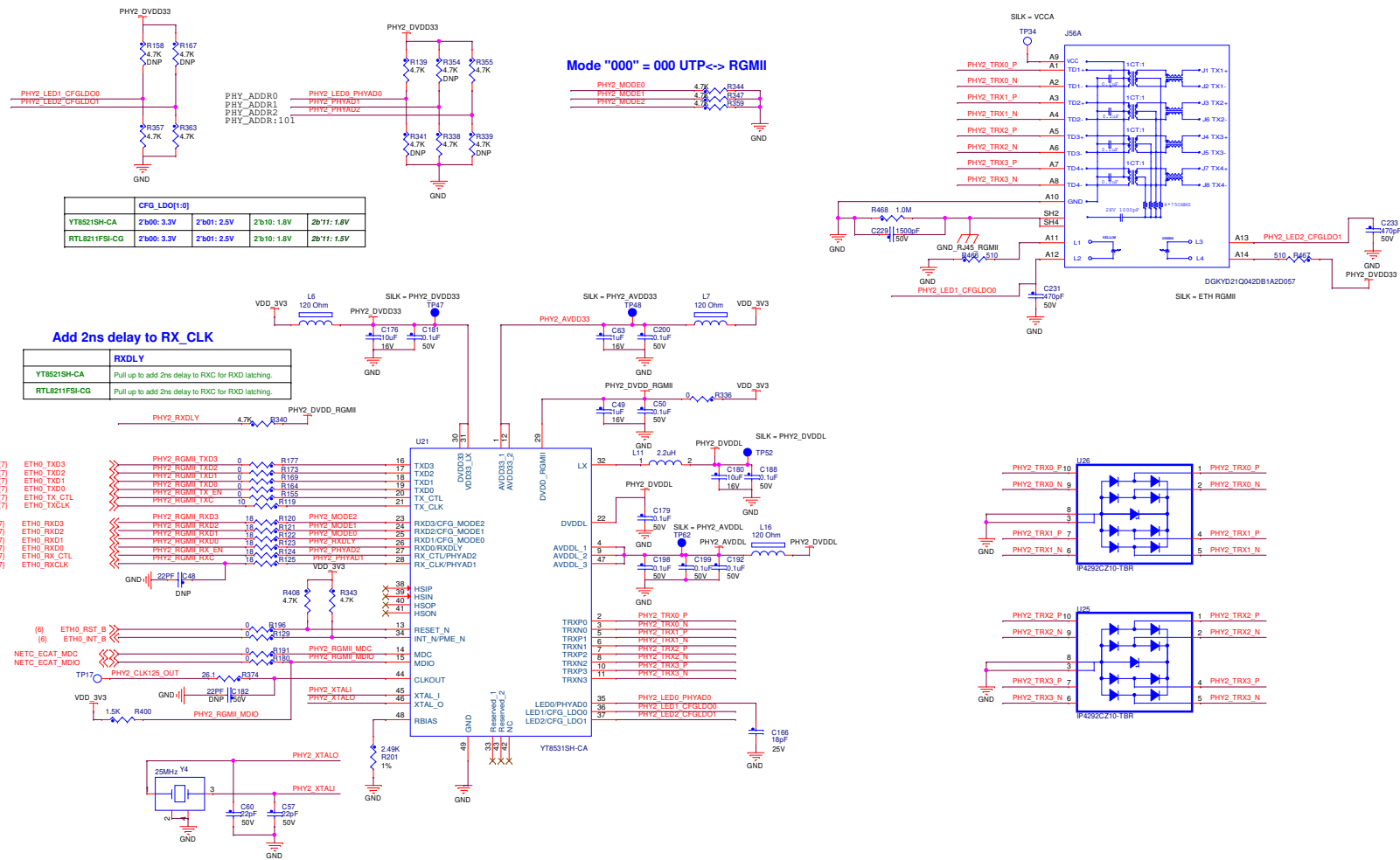


MII Interface

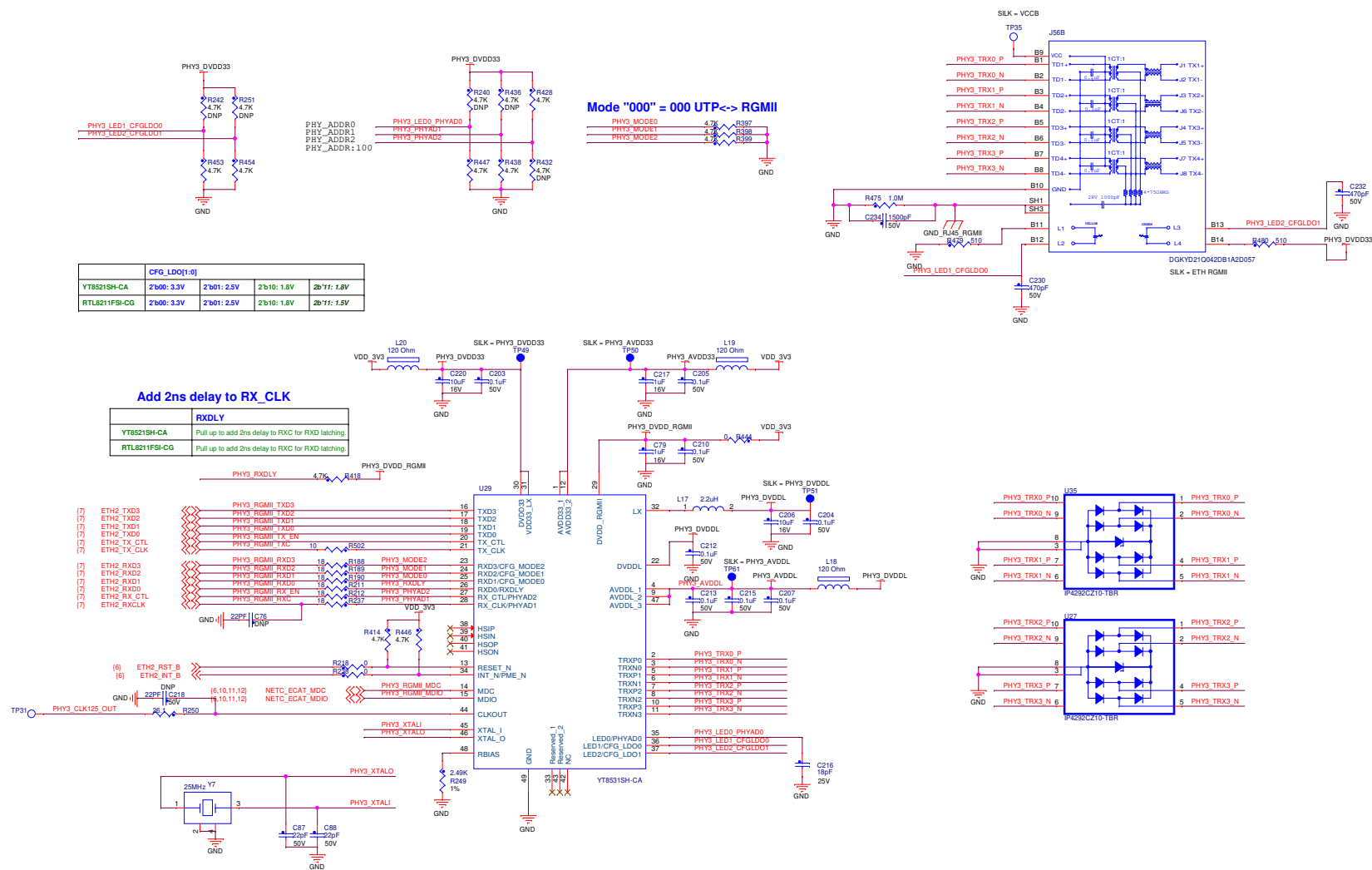


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Size C	Document Number	SCH-95302, PDF-SPF-95302	
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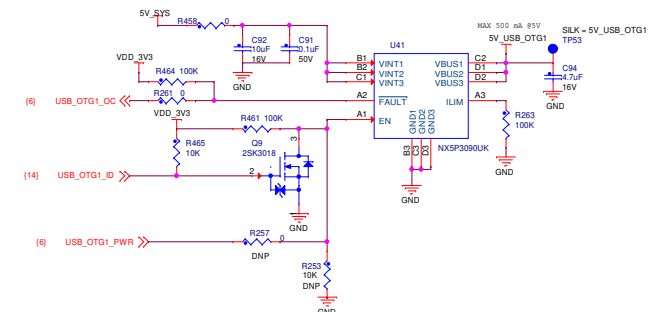
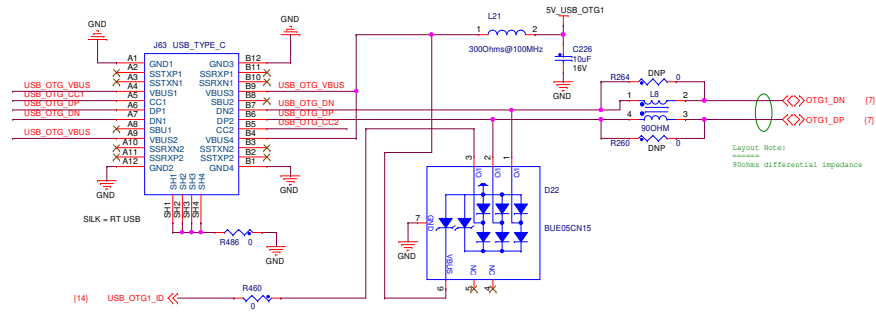
RGMII ETH0 {J12(1-2)/J13(2-3): ETH0 Function Via J56A}



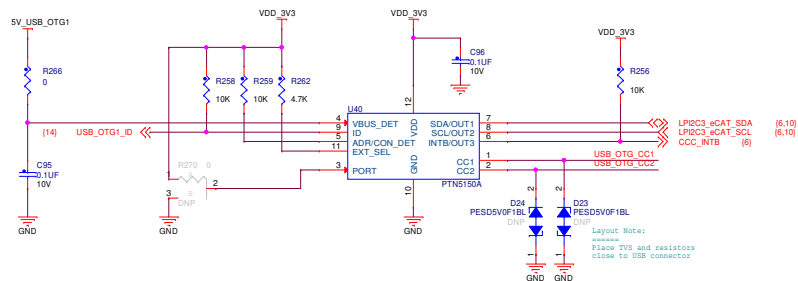
RGMII ETH2 {J18 (1-2) / J17 (2-3) : ETH2 Function Via J56B}



USB OTG1



CC Logic



EXT_SEL

0---CC2
1---CC1

Default

1

PORT

PORT=1: DFP mode
PORT=0: UFP mode
PORT=Float: DRP mode

PORT=Float: DRP mode

ADR/CON_DET

CON_DET=1, Connection Detected, I2C Address: 0x7A
CON_DET=0, No Connection, I2C Address: 0x3A

Float, no I2C function

USB ID is a processor Analog Sense Input

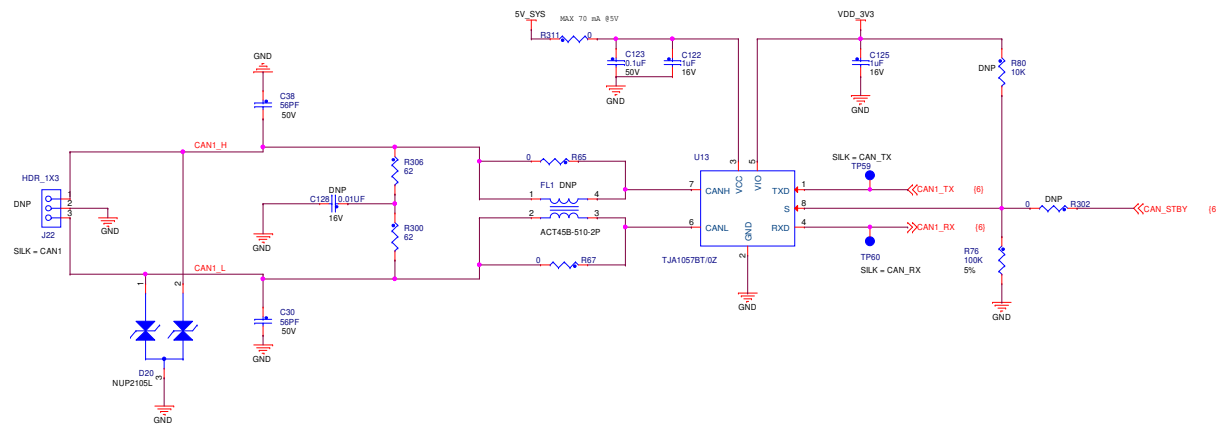
Host --> ID = GND

Device --> ID = Floating



Classification:	<FCP>	<PUB>
Drawing Title:	FRDM-IMXRT1186	
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CAN BUS



Classification:	<FCP>	<PUB>
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Drawing Title: **FRDM-IMXRT1186**

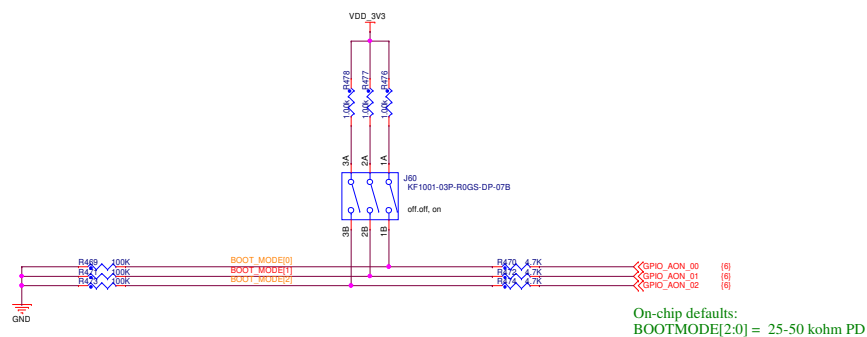
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Size C	Document Number SCH-95302, PDF:SPF-95302
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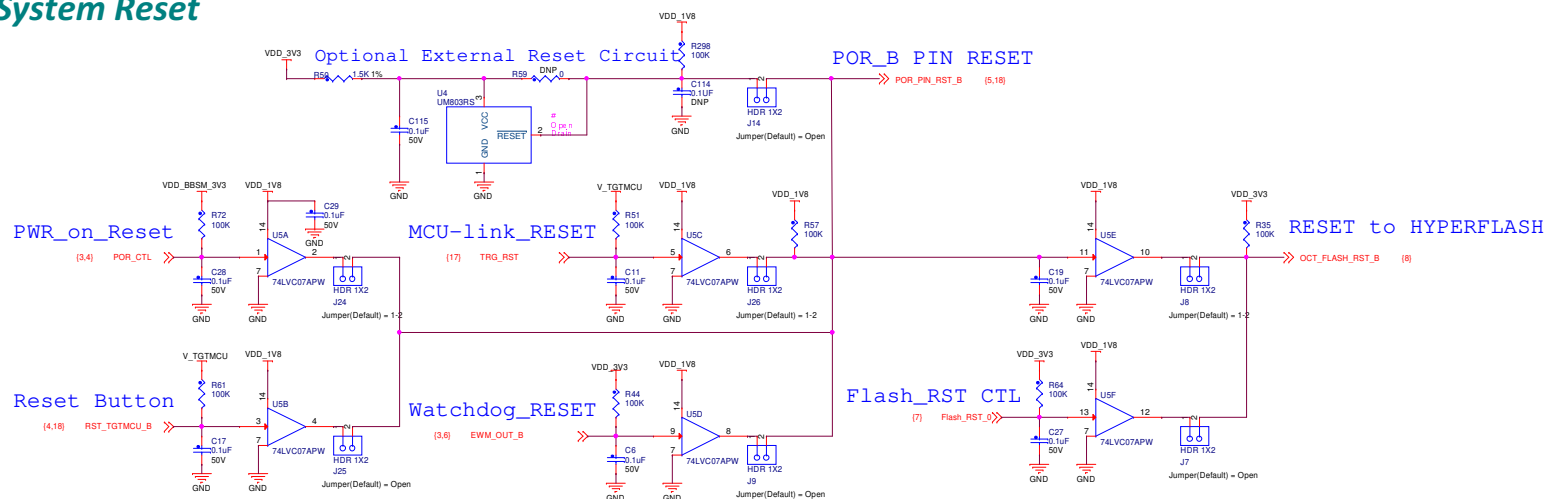
External Boot Switch



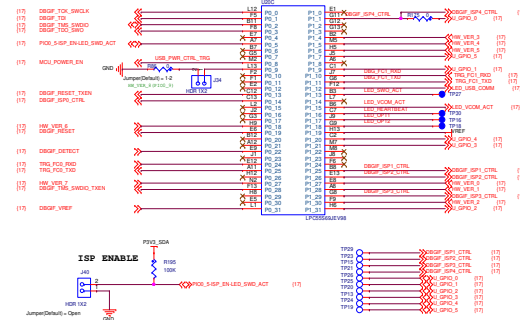
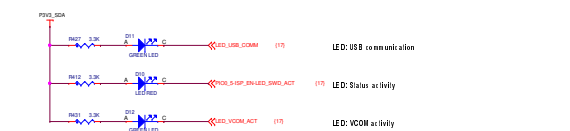
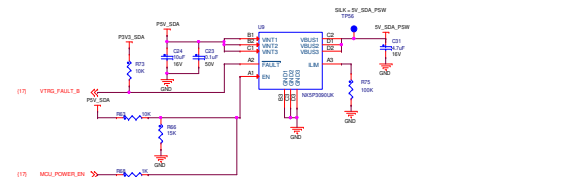
Boot Mode pin settings

BOOT_MODE[2:0]	Boot Type
000(J60-x000)	Boot from Internal Fuses
001(J60-x001)	Serial Downloader (USB1 or LPUART1)
010(J60-x010)	USDHC2 8-bit 1.8V eMMC 5.1
011(J60-x011)	uSDHC1 4-bit SD 3.0
100(J60-x100)	FlexSPI Quad SPI Serial NOR (or Octal SPI NOR in Quad SPI NOR mode)
101(J60-x101)	FlexSPI Quad SPI Serial NAND 2k page (or Octal SPI NAND in Quad SPI NOR mode)
110(J60-x110)	FlexSPI Quad SPI Serial NAND 4k page (or Octal SPI NAND in Quad SPI NOR mode)
111(J60-x111)	Test Mode

System Reset



Classification:		<FCP>	<PUB>
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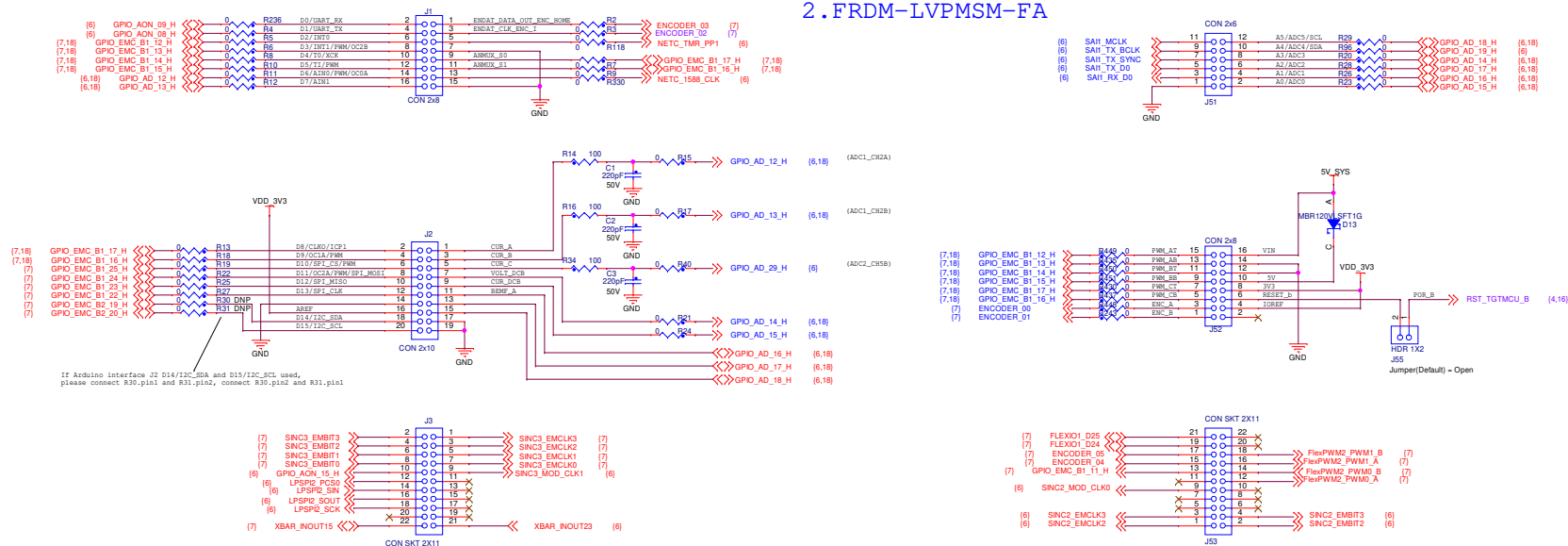
[illegible]

Items	Description	OB setting
HW_VER_0(PIO1_27)	Power negotiation when low	0
HW_VER_1(PIO1_28)	Reserved	1
HW_VER_2(PIO1_30)	USBIO bridge disabled / not present when 0	0
HW_VER_3(PIO1_4)	OB (0 / Pro (1) type select	0
HW_VER_4(PIO1_5)	Board identity code used (when low)	0
HW_VER_5(PIO1_6)	Power measurement enabled/present when low	1
HW_VER_6(PIO1_18)	VCOM disabled when low	1
HW_VER_7(PIO0_27)	SWD debug disabled when low	1
HW_VER_8(PIO0_9)	Secondary VCOM disabled when low.	0

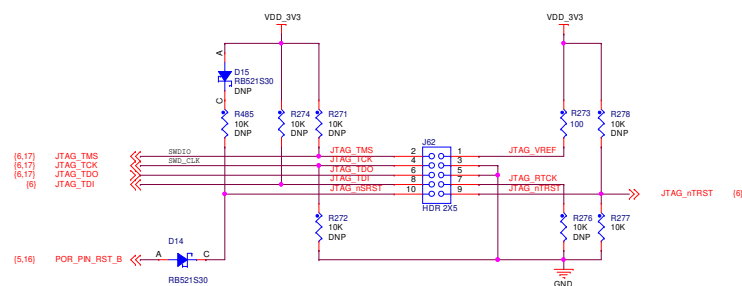
Arduino & MC Interface

Motor Card Supported:

- 1.FRDM-MC-LVPMSM
- 2.FRDM-LVPMSM-FA

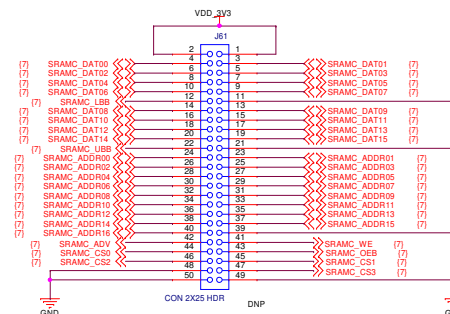


JTAG(SWD)



SRAMC Interface (Reserved)

To Enable SRAMC:
Related Jumper Wiring Change to 1-3



Classification: <FCP>		<PUBB>	
Drawing Title: FRDM-IMXRT1186			
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